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United States Patent	12412875
Kind Code	B2
Date of Patent	September 09, 2025
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Semiconductor package

Abstract

A semiconductor package includes a substrate having a passive element region, a peripheral region adjacent to the passive element region, and a remaining region, a first passive element on an upper surface of the passive element region, a first semiconductor chip on an upper surface of the remaining region, and a sealing portion covering the substrate, the first passive element, and the first semiconductor chip, wherein the peripheral region includes a first sub-region on a first side of the first passive element, a second sub-region on a second side opposite the first side, a third sub-region on a third side of the first passive element, and a fourth sub-region on a fourth side opposite the third side, and wherein a roughness of an upper surface of at least one of the first to fourth sub-regions is greater than a roughness of the upper surface of the remaining region.

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Appl. No.: **17/725729**

Filed: **April 21, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230111555 A1	Apr. 13, 2023

Foreign Application Priority Data

KR	10-2021-0134436	Oct. 08, 2021

Publication Classification

Int. Cl.: **H01L25/16** (20230101); H01L23/00 (20060101); H01L23/498 (20060101)

U.S. Cl.:

CPC **H01L25/16** (20130101); H01L23/49816 (20130101); H01L23/49822 (20130101); H01L23/49838 (20130101); H01L24/16 (20130101); H01L24/48 (20130101); H01L2224/16227 (20130101); H01L2224/16237 (20130101); H01L2224/48228 (20130101); H01L2924/1431 (20130101); H01L2924/1436 (20130101); H01L2924/1438 (20130101)

Field of Classification Search

CPC: H01L (2224/16111); H01L (2224/16225); H01L (2224/48091); H01L (2224/48227); H01L (2224/16227); H01L (2224/16237); H01L (2224/48228); H01L (2224/32013); H01L (2224/32054); H01L (2224/32059); H01L (2224/32104); H01L (2924/00014); H01L (2924/15192); H01L (2924/15311); H01L (2924/181); H01L (2924/19105); H01L (2924/1431); H01L (2924/1436); H01L (2924/1438); H01L (24/32); H01L (24/16); H01L (24/48); H01L (24/13); H01L (23/3128); H01L (23/49816); H01L (23/49822); H01L (23/49838); H01L (23/50); H01L (25/16); H01L (25/0655); H01L (25/0657)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0134436, filed on Oct. 8, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(2) Embodiments relate to a semiconductor package. In particular, embodiments relate to a semiconductor package including a passive element and a semiconductor chip.

2. Description of the Related Art

(3) A solder paste may be used when a passive element is mounted on a package substrate. A passive element connecting member between the package substrate and the passive element and a flux layer on the package substrate may be made from the solder paste.

SUMMARY

(4) According to an aspect of embodiments, there is provided a semiconductor package that may include a package substrate having a passive element region, a passive element adjacent region adjacent to the passive element region, and a remaining region; a first passive element on an upper surface of the passive element region; a first semiconductor chip on an upper surface of the remaining region; and a sealing portion covering the package substrate, the first passive element, and the first semiconductor chip, wherein the passive element adjacent region includes a first sub-region on a first side of the first passive element, a second sub-region on a second side opposite the first side of the first passive element, a third sub-region on a third side of the first passive element, and a fourth sub-region on a fourth side opposite the third side of the first passive element, and a roughness of an upper surface of at least one of the first sub-region to the fourth sub-region is greater than the roughness of an upper surface of the remaining region.

(5) According to another aspect of embodiments, there is provided a semiconductor package that may include a package substrate having a wire pad, a wire pad, a wire pad adjacent region adjacent to the wire pad, and a remaining region; a passive element on an upper surface of the remaining region; a semiconductor chip on the upper surface of the remaining region; a wire electrically connecting the semiconductor chip to the wire pad; and a sealing portion covering the package substrate, the passive element, the semiconductor chip, and the wire, wherein the roughness of an upper surface of the wire pad adjacent region is less than the roughness of the upper surface of the remaining region.

(6) According to another aspect of embodiments, there is provided a semiconductor package that may include a package substrate having a passive element region, a passive element adjacent region adjacent to the passive element region, and a remaining region; a plurality of external connection terminals on a lower surface of the package substrate; a passive element on an upper surface of the passive element region; a memory controller chip on the upper surface of the remaining region; a chip bump connecting the memory controller chip to the package substrate; a memory chip on the memory controller chip or the remaining region; a wire connecting the memory chip to the package substrate; and a sealing portion covering the package substrate, the passive element, the memory controller chip, the chip bump, the memory chip, and the wire, wherein the passive element adjacent region includes a first sub-region on a first side of the first passive element, a second sub-region on a second side opposite the first side of the first passive element, a third sub-region on a third side of the first passive element, and a fourth sub-region on a fourth side opposite the third side of the first passive element, and the roughness of the upper surface of at least one of the first sub-region to the fourth sub-region is greater than the roughness of the upper surface of the remaining region.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features will become apparent to those of skill in the art by describing in detail exemplary embodiments with reference to the attached drawings, in which:

(2) FIG. **1**A is a cross-sectional view of a semiconductor package according to an embodiment;

(3) FIG. **1**B is an enlarged view of region B in FIG. **1**A;

(4) FIG. **1**C is a plan view of a semiconductor package according to an embodiment;

(5) FIG. **2** is a cross-sectional view of a semiconductor package according to an embodiment;

(6) FIG. **3** is a plan view of a semiconductor package according to an embodiment;

(7) FIG. **4** is a plan view of a semiconductor package according to an embodiment;

(8) FIG. **5** is a plan view of a semiconductor package according to an embodiment;

(9) FIG. **6** is a plan view of a semiconductor package according to an embodiment;

(10) FIG. **7** is a plan view of a semiconductor package according to an embodiment;

(11) FIG. **8** is a plan view of a semiconductor package according to an embodiment;

(12) FIG. **9** is a plan view of a semiconductor package according to one embodiment;

(13) FIG. **10** is a plan view of a semiconductor package according to an embodiment;

(14) FIG. **11** is a plan view of a semiconductor package according to an embodiment;

(15) FIG. **12** is a plan view of a semiconductor package according to an embodiment;

(16) FIG. **13** is a plan view of a semiconductor package according to an embodiment;

(17) FIG. **14** is a plan view of a semiconductor package according to an embodiment;

(18) FIG. **15** is a cross-sectional view of a semiconductor package according to an embodiment;

(19) FIG. **16** is a cross-sectional view of a semiconductor package according to an embodiment;

(20) FIG. **17** is a cross-sectional view of a semiconductor package according to an embodiment;

(21) FIG. **18** is a cross-sectional view of a semiconductor package according to an embodiment;

(22) FIG. **19** is a cross-sectional view of a semiconductor package according to an embodiment;

(23) FIG. **20**A is a cross-sectional view of a semiconductor package according to an embodiment;

(24) FIG. **20**B is an enlarged view of region B**2** in FIG. **20**A; and

(25) FIG. **21** is a flowchart of a method of manufacturing a semiconductor package, according to an embodiment.

DETAILED DESCRIPTION

(26) FIG. **1**A is a cross-sectional view showing a semiconductor package **1000** according to an embodiment. FIG. **1**B is an enlarged view of region B in FIG. **1**A, and FIG. **1**C is a plan view showing the semiconductor package **1000** according to an embodiment.

(27) Referring to FIGS. **1**A to **1**C, the semiconductor package **1000** may include a package substrate **100**, a plurality of external connection terminals **500** on a lower surface of the package substrate **100**, a passive element **200** on an upper surface of the package substrate **100**, a plurality of passive element connecting members **250** for connecting the passive element **200** to the package substrate **100**, a semiconductor chip **300** on the upper surface of the package substrate **100**, a plurality of chip bumps **350** connecting the semiconductor chip **300** to the package substrate **100**, and a sealing portion **400** covering the semiconductor chip **300**. In some embodiments, the semiconductor package **1000** may further include a flux layer **270** on the package substrate **100**.

(28) The package substrate **100** may include a printed circuit board (PCB). For example, the package substrate **100** may include a core layer **111**, a first insulating layer **121***a* on an upper surface of the core layer **111**, a second insulating layer **121***b* on a lower surface of the core layer **111**, a third insulating layer **121***c* on the upper surface of the first insulating layer **121***a*, a fourth insulating layer **121***d* on the lower surface of the second insulating layer **121***b*, a fifth insulating layer **121***e* on the upper surface of the third insulating layer **121***c*, and a sixth insulating layer **121***f* on the lower surface of the fourth insulating layer **121***d*. The package substrate **100** may further

include a first protective layer **130** on an upper surface of the fifth insulating layer **121***e* and a second protective layer **140** on a lower surface of the sixth insulating layer **121***f*.
(29) The package substrate **100** may further include a core via **113** penetrating the core layer **111**. The package substrate **100** may further include a first core wiring layer **112***a* positioned on the upper surface of the core layer **111** and in contact with the core via **113**. The package substrate **100** may further include a second core wiring layer **112***b* positioned on the lower surface of the core layer **111** and in contact with the core via **113**.
(30) The package substrate **100** may further include a first wiring layer **122***a* on the upper surface of the first insulating layer **121***a*. The package substrate **100** may further include a first via layer **123***a* passing through the first insulating layer **121***a* between the first wiring layer **122***a* and the first core wiring layer **112***a*. The package substrate **100** may further include a second wiring layer **122***b* on a lower surface of the second insulating layer **121***b*. The package substrate **100** may further include a second via layer **123***b* passing through the second insulating layer **121***b* between the second wiring layer **122***b* and the second core wiring layer **112***b*. The package substrate **100** may further include a third wiring layer **122***c* on the upper surface of the third insulating layer **121***c*. The package substrate **100** may further include a third via layer **123***c* passing through the third insulating layer **121***c* between the third wiring layer **122***c* and the first wiring layer **122***a*. The package substrate **100** may further include a fourth wiring layer **122***d* on the lower surface of the fourth insulating layer **121***d*. The package substrate **100** may further include a fourth via layer **123***d* passing through the fourth insulating layer **121***d* between the fourth wiring layer **122***d* and the second wiring layer **122***b*.
(31) The package substrate **100** may further include a plurality of bump pads **124***a***-1** and a plurality of passive element pads **124***a***-2** on the upper surface of the fifth insulating layer **121***e*. The first protective layer **130** may have a plurality of openings exposing the plurality of bump pads **124***a***-1** and the plurality of passive element pads **124***a***-2**, respectively. The package substrate **100** may further include a fifth via layer **123***e* passing through the fifth insulating layer **121***e* between the third wiring layer **122***c* and the plurality of bump pads **124***a***-1** and between the third wiring layer **122***c* and the plurality of passive element pads **124***a***-2**. The package substrate **100** may further include a plurality of external connection terminal pads **124***b* on the lower surface of the sixth insulating layer **121***f*. The second protective layer **140** may have a plurality of openings exposing the plurality of external connection terminal pads **124***b*. The package substrate **100** may further include a sixth via layer **123***f* passing through the sixth insulating layer **121***f* between the fourth wiring layer **122***d* and the plurality of external connection terminal pads **124***b*.
(32) In FIG. **1**A, the package substrate **100** includes six insulating layers (i.e., the first to sixth insulating layers **121***a* to **121***f*), four wiring layers (i.e., the first to fourth wiring layers **122***a* to **122***d*), and six via layers (i.e., the first to sixth via layers **123***a* to **123***f*). However, the numbers of insulating layers, wiring layers, and via layers included in the package substrate **100** may be variously modified.
(33) For example, the core layer **111** may include an insulating material, a thermosetting resin (e.g., an epoxy resin) or a thermoplastic resin (e.g., polyimide). The core layer **111** may include a material including a reinforcing material, e.g., fiberglass, and/or an inorganic filler, e.g., a copper clad laminate (CCL) or an unclad CCL. The core layer **111** may include a metal plate, a glass plate, or a ceramic plate.
(34) The first to sixth insulating layers **121***a* to **121***f* may include a thermosetting resin, e.g., epoxy, or a thermoplastic resin, e.g., polyimide. In some embodiments, the first to sixth insulating layers **121***a* to **121***f* may include a material including a reinforcing material, e.g., glass fiber, and/or an inorganic filler in addition to a thermoplastic resin and/or a thermosetting resin, e.g., prepreg or Ajinomoto build-up film (ABF). In some embodiments, the first protective layer **130** and the second protective layer **140** may include solder resist.
(35) The core via **113**, the first and second core wiring layers **112***a* and **112***b*, the first to fourth

wiring layers **122***a* to **122***d*, the first to sixth via layers **123***a* to **123***f*, the bump pads **124***a***-1**, the passive element pads **124***a***-2**, and the external connection terminal pads **124***b* may include a metal material. The metal material may include, e.g., copper (Cu), aluminum (Al), silver (Ag), tin (Sn), gold (Au), nickel (Ni), lead (Pb), titanium (Ti), or an alloy thereof.

(36) As shown in FIG. **1**C, the package substrate **100** may have a passive element region R**1**, a passive element adjacent region R**2** adjacent to the passive element region R**1**, and a remaining region R**3**. For example, the passive element adjacent region R**2** may be a peripheral region of the passive element region R**1**. In some embodiments, the passive element adjacent region R**2** may surround, e.g., an entire perimeter of, the passive element region R**1**, and the remaining region R**3** may surround, e.g., an entire perimeter of, the passive element adjacent region R**2**, as viewed in a top view (FIG. **1**C).

(37) The plurality of external connection terminals **500** may be respectively positioned on the plurality of external connection terminal pads **124***b* of the package substrate **100**. Each of the external connection terminals **500** may include, e.g., tin (Sn) or a tin (Sn) alloy. In some embodiments, each external connection terminal **500** may be formed of a solder ball.

(38) The passive element **200** may be located on the passive element region R**1**. The passive element **200** may be positioned on the plurality of passive element pads **124***a***-2** of the package substrate **100**. The passive element **200** may be, e.g., a capacitor, an inductor, or a resistor.

(39) The plurality of passive element connecting members **250** may be located between the passive element region R**1** of the package substrate **100** and the passive element **200**, e.g., the plurality of passive element connecting members **250** may be between the package substrate **100** and the passive element **200** along the Z direction. The plurality of passive element connecting members **250** may be located between the passive element **200** and the plurality of passive element pads **124***a***-2**, respectively. The plurality of passive element connecting members **250** may include, e.g., tin (Sn) or tin (Sn) alloys. In some embodiments, a plurality of passive element connecting members **250** may be formed of a solder paste.

(40) The semiconductor chip **300** may be located on the remaining region R**3**. The semiconductor chip **300** may be located on the plurality of bump pads **124***a***-1** of the package substrate **100**. The semiconductor chip **300** may include an arbitrary kind of integrated circuit including, e.g., a memory circuit, a logic circuit, or a combination thereof. The memory circuit may be, e.g., a dynamic random access memory (DRAM) circuit, a static random access memory (SRAM) circuit, a flash memory circuit, an electrically erasable and programmable read-only memory (EEPROM) circuit, a phase-change random access memory (PRAM) circuit, a magnetic random access memory (MRAM) circuit, a resistive random access memory (RRAM) circuit, or a combination thereof. The logic circuit may be, e.g., a central processing unit (CPU) circuit, a graphics processing unit (GPU) circuit, a memory controller circuit, an application specific integrated circuit (ASIC) circuit, an application processor (AP) circuit, or a combination thereof.

(41) The chip bumps **350** may be located between the remaining region R**3** of the package substrate **100** and the semiconductor chip **300**, e.g., the chip bumps **350** may be between the package substrate **100** and the semiconductor chip **300** along the Z direction. The chip bumps **350** may be located between the semiconductor chip **300** and the plurality of bump pads **124***a***-1**, respectively. The chip bumps **350** may include, e.g., a tin (Sn) or a tin (Sn) alloy. In some embodiments, the chip bump **350** may be formed of a solder ball.

(42) The sealing portion **400** may cover the package substrate **100**, the passive element **200**, and the semiconductor chip **300**. The sealing portion **400** may include an epoxy resin. For example, the sealing portion **400** may include an epoxy mold compound (EMC).

(43) The flux layer **270** may be located on the passive element adjacent region R**2**. In some embodiments, the flux layer **270** may be further positioned on the passive element region R**1**. In some embodiments, the flux layer **270** may be formed of flux of solder paste.

(44) Referring to FIG. **1**C, the passive element adjacent region R**2** may include a first sub-region

R**2***a* on the first side of the passive element **200**, a second sub-region R**2***b* on a second side opposite to the first side of the passive element **200**, a third sub-region R**2***c* on a third side of the passive element **200**, and a fourth sub-region R**2***d* of a fourth side opposite to the third side of the passive element **200**. In some embodiments, the passive element adjacent region R**2** may further include a fifth sub-region R**2***e* in which the first sub-region R**2***a* intersects with the third sub-region R**2***c*, a sixth sub-region R**2***f* in which the first sub-region R**2***a* intersects with the fourth sub-region R**2***d*, a seventh sub-region R**2***g* in which the third sub-region R**2***c* intersects with the second sub-region R**2***b*, and an eighth sub-region R**2***h* in which the fourth sub-region R**2***d* intersects with the second sub-region R**2***b*. In some embodiments, e.g., each of, the width Wa of the first sub-region R**2***a*, the width Wb of the second sub-region R**2***b*, the width Wc of the third sub-region R**2***c*, and the width Wd of the fourth sub-region R**2***d* may be about 10 μm to about 500 μm.
(45) In some embodiments, the roughness of the upper surface of the first sub-region R**2***a* to the fourth sub-region R**2***d* may be greater than the roughness of the upper surface of the remaining region R**3**. For example, referring to FIG. **1**B, the roughness of the upper surface, i.e., a surface facing the passive element **200**, of the first sub-region R**2***a* to the fourth sub-region R**2***d* may be greater than the roughness of the upper surface, i.e., a surface facing the semiconductor chip **300**, of the remaining region R**3**.
(46) In this specification, roughness means average roughness (roughness average, Ra). In some embodiments, the roughness of the upper surface of the first sub-region R**2***a* to the fourth sub-region R**2***d* may be greater than the roughness of the upper surface of the remaining region R**3** by about 80 nm or more. In some embodiments, the roughness of the upper surface of the fifth sub-region R**2***e* to the eighth sub-region R**2***h* may be greater than the roughness of the upper surface of the remaining region R**3**. In some embodiments, the roughness of the upper surface of the fifth sub-region R**2***e* to the eighth sub-region R**2***h* may be greater than the roughness of the upper surface of the remaining region R**3** by about 80 nm or more.
(47) The greater the roughness of the surface through which the flux flows, the wider the flux may flow. Because the passive element adjacent region R**2** has a rough surface, the flux layer **270** may be formed wider on the passive element adjacent region R**2**. Accordingly, the concentration of the flux mixed in the sealing portion **400** may be reduced. Therefore, the possibility of delamination caused by combining the sealing portion **400** with the flux layer **270** may be reduced. On the other hand, by forming the remaining region R**3** to be relatively flat, i.e., a lower surface roughness, the flow of the flux layer **270** to the remaining region R**3** may be reduced. Therefore, the flux layer **270** may be prevented from contaminating the bump pads **124***a***-1** located in the remaining region R**3**.
(48) In some embodiments, the roughness of the upper surface of the passive element region R**1** may be greater than the roughness of the upper surface of the remaining region R**3**. For example, the roughness of the upper surface of the passive element region R**1** may be greater than the roughness of the upper surface of the remaining region R**3** by about 80 nm or more.
(49) FIG. **2** is a cross-sectional view showing a semiconductor package **1000***a* according to an embodiment. Hereinafter, differences between the semiconductor package **1000** shown in FIG. **1**A to **1**C and the semiconductor package **1000***a* shown in FIG. **2** are described.
(50) Referring to FIG. **2**, the roughness of the upper surface of the passive element region R**1** may be less than the roughness of the upper surface of the passive element adjacent region R**2**. For example, the roughness of the upper surface of the passive element region R**1** may be less than the roughness of the upper surface of the first sub-region R**2***a*. For example, the roughness of the upper surface of the passive element region R**1** may be less than the roughness of the upper surface of the passive element adjacent region R**2** by about 80 nm or more. The roughness of the upper surface of the passive element region R**1** may be less than the roughness of the upper surface of the second to fourth sub regions (refer to R**2***b* to R**2***d*, and FIG. **1**C). For example, the roughness of the upper surface of the passive element region R**1** may be less than the roughness of the upper surface of the upper surface of the second to fourth sub regions R**2***b* to R**2***d* (refer to FIG. **1**C) by about 80 nm or

more.

(51) FIG. **3** is a plan view showing a semiconductor package **1000***b* according to an embodiment. Hereinafter, differences between the semiconductor package **1000** shown in FIGS. **1**A to **1**C and the semiconductor package **1000***b* shown in FIG. **3** are described.

(52) Referring to FIG. **3**, the roughness of an upper surface of a first sub-region R**2***a*-**1** may be less than the roughness of an upper surface of a second sub-region R**2***b* to a fourth sub-region R**2***d*. For example, the roughness of the upper surface of the first sub-region R**2***a*-**1** may be less than the roughness of the upper surface of the second sub-region R**2***b* to the fourth sub-region R**2***d* by 80 nm or more.

(53) In some embodiments, the roughness of upper surfaces of a fifth sub-region R**2***e*-**1** and a sixth sub-region R**2***f*-**1** may be less than roughness of the upper surfaces of a seventh sub-region R**2***g* and an eighth sub-region R**2***h*. For example, the roughness of the upper surface of the fifth sub-region R**2***e*-**1** and the sixth sub-region R**2***f*-**1** may be less than the roughness of the upper surface of the seventh sub-region R**2***g* and the eighth sub-region R**2***h* by 80 nm or more

(54) By forming the first sub-region R**2***a*-**1**, the fifth sub-region R**2***e*-**1**, and the sixth sub-region R**2***f*-**1** to be relatively flat, the flow of the flux layer **270** to the first side of the passive element **200** may be reduced. Accordingly, the flux layer **270** may be prevented from contaminating the bump pad **124***a*-**1** (refer to FIG. **1**A) under the semiconductor chip **300** positioned on the first side of the passive element **200**.

(55) FIG. **4** is a plan view showing a semiconductor package **1000***c* according to an embodiment. Hereinafter, differences between the semiconductor packages **1000** shown in FIGS. **1**A to **1**C and the semiconductor package **1000***c* shown in FIG. **4** are described.

(56) Referring to FIG. **4**, the semiconductor package **1000***c* may include the package substrate **100**, a first passive element **201**, a first passive element connecting member **251**, a first flux layer **271**, a second passive element **202**, a second passive element connecting member **252**, a second flux layer **272**, and the semiconductor chip **300**. The package substrate **100** may include a first passive element region R**1***a*, a first passive element adjacent region R**2** adjacent to the first passive element region R**1***a*, a second passive element region R**1***b* positioned on the first side of the first passive element region R**1***a*, a second passive element adjacent region R**4** adjacent to the second passive element region R**1***b*, and a remaining region R**3**. The first passive element **201** may be located on the first passive element region R**1***a*, and the second passive element **202** may be located on the second passive element region R**1***b*, and the semiconductor chip **300** may be located on the remaining region R**3**. A plurality of first passive element connecting members **251** may connect the first passive element **201** to the package substrate **100**, and a plurality of second passive element connecting members **252** may connect the second passive element **202** to the package substrate **100**. The first flux layer **271** may be located on the first passive element region R**1***a* and the first passive element adjacent region R**2**, and the second flux layer **272** may be located on the second passive element region R**1***b* and the second passive element adjacent region R**4**.

(57) The first passive element adjacent region R**2** may include a first sub-region R**2***a*-**1** on a first side of the first passive element **201**, a second sub-region R**2***b* on a second side opposite to the first side of the first passive element **201**, a third sub-region R**2***c* on a third side of the first passive element **201**, and the fourth sub-region R**2***d* on a fourth side opposite to the third side of the first passive element **201**. In some embodiments, the first passive element adjacent region R**2** may further include a fifth sub-region R**2***e*-**1** in which the first sub-region R**2***a*-**1** intersects with the third sub-region R**2***c*, a sixth sub-region R**2***f*-**1** in which the first sub-region R**2***a*-**1** intersects with the fourth sub-region R**2***d*, a seventh sub-region R**2***g* in which the third sub-region R**2***c* intersects with the second sub-region R**2***b*, and an eighth sub-region R**2***h* in which the fourth sub-region R**2***d* intersects with the second sub-region R**2***b*.

(58) The second passive element **202** may be located on the first side of the first passive element **201**, e.g., the second passive element **202** may be between the first passive element **201** and the

semiconductor chip **300** along the X direction. In this case, the roughness of the upper surface of the first sub-region R**2***a***-1** of the first passive element adjacent region R**2** may be less than the roughness of the upper surfaces of the second sub-region R**2***b* to the fourth sub-region R**2***d* of the first passive element adjacent region R**2**. For example, the roughness of the upper surface of the first sub-region R**2***a***-1** of the first passive element adjacent region R**2** may be less than that of the upper surface of the second sub-region R**2***b* to the fourth sub-region R**2***d* of the first passive element adjacent region R**2** by 80 nm or more.
(59) The roughness of the upper surfaces of the fifth sub-region R**2***e***-1** and the sixth sub-region R**2***f***-1** of the first passive element adjacent region R**2** may be less than the roughness of the upper surfaces of the seventh sub-region R**2***g* and the eighth sub-region R**2***h* of the first passive element adjacent region R**2**. For example, the roughness of upper surfaces of the fifth sub-region R**2***e***-1** and the sixth sub-region R**2***f***-1** of the first passive element adjacent region R**2** may be less than the roughness of upper surfaces of the seventh sub-region R**2***g* and the eighth sub-region R**2***h* of the first passive element adjacent region R**2** by 80 nm or more.
(60) By forming the first sub-region R**2***a***-1**, the fifth sub-region R**2***e***-1**, and the sixth sub-region R**2***f***-1** of the first passive element adjacent region R**2** relatively flat, the flow of the first flux layer **271** to the first side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting, e.g., contacting, the second flux layer **272**. Therefore, it is possible to prevent an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. **1**A). Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. **1**A) and the package substrate **100**.
(61) The second passive element adjacent region R**4** may include a first sub-region R**4***a***-1** on a first side of the second passive element **202**, a second sub-region R**4***b***-1** on a second side opposite to the first side of the second passive element **202**, a third sub-region R**4***c* on a third side of the second passive element **202**, and a fourth sub-region R**4***d* on a fourth side opposite to the third side of the second passive element **202**. In some embodiments, the second passive element adjacent region R**4** may further include a fifth sub-region R**4***e***-1** in which the first sub-region R**4***a***-1** intersects with the third sub-region R**4***c*, a sixth sub-region R**4***f***-1** in which the first sub-region R**4***a***-1** intersects with the fourth sub-region R**4***d*, a seventh sub-region R**4***g***-1** in which the third sub-region R**4***c* intersects with the second sub-region R**4***b***-1**, and an eighth sub-region R**4***h***-1** in which the fourth sub-region R**4***d* intersects with the second sub-region R**4***b***-1**.
(62) The first passive element **201** may be positioned on the second side of the second passive element **202**, and the semiconductor chip **300** may be positioned on the first side of the second passive element **202**. In this case, the roughness of the upper surfaces of the first sub-region R**4***a***-1** and the second sub-region R**4***b***-1** of the second passive element adjacent region R**4** may be less than the roughness of the upper surfaces of the third sub-region R**4***c* and the fourth sub-region R**4***d* of the second passive element adjacent region R**4**. For example, the roughness of upper surfaces of the first sub-region R**4***a***-1** and the second sub-region R**4***b***-1** of the second passive element adjacent region R**4** may be less than the roughness of the upper surfaces of the third sub-region R**4***c* and the fourth sub-region R**4***d* of the second passive element adjacent region R**4** by 80 nm or more.
(63) The roughness of upper surfaces of the fifth sub-region R**4***e***-1** to the eighth sub-region R**4***h***-1** of the second passive element adjacent region R**4** may be less than the roughness of the upper surfaces of the third sub-region R**4***c* and the fourth sub-region R**4***d* of the second passive element adjacent region R**4**. For example, the roughness of upper surfaces of the fifth sub-region R**4***e***-1** to the eighth sub-region R**4***h***-1** of the second passive element adjacent region R**4** may less than the roughness of the upper surfaces of the third sub-region R**4***c* and the fourth sub-region R**4***d* of the second passive element adjacent region R**4** by 80 nm or more.
(64) By forming the first sub-region R**4***a***-1**, the fifth sub-region R**4***e***-1**, and the sixth sub-region R**4***f***-1** of the second passive element adjacent region R**4** to be relatively flat, the flow of the second flux layer **272** to the first side of the second passive element **202** may be reduced. Accordingly, the

second flux layer **272** may be prevented from contaminating the bump pad **124***a***-1** (refer to FIG. **1**A) under the semiconductor chip **300** positioned on the first side of the second passive element **202**.
(65) By forming the second sub-region R**4***b***-1**, the seventh sub-region R**4***g***-1**, and the eighth sub-region R**4***h***-1** of the second passive element adjacent region R**4** to be relatively flat, the flow of the second flux layer **272** to the second side of the second passive element **202** may be reduced. Accordingly, the second flux layer **272** may be prevented from meeting, e.g., contacting, the first flux layer **271**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. **1**A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. **1**A) and the package substrate **100**.
(66) FIG. **5** is a plan view showing a semiconductor package **1000***d* according to an embodiment. Hereinafter, differences between the semiconductor package **1000***c* shown in FIG. **4** and the semiconductor package **1000***d* shown in FIG. **5** are described.
(67) Referring to FIG. **5**, the semiconductor chip **300** may be positioned on a third side of the first passive element **201**, e.g., the semiconductor chip **300** and the first passive element **201** may be adjacent to each other along the Y direction, and the second passive element **202** may be positioned on a first side of the first passive element **201**, e.g., the first and second passive elements **201** and **202** may adjacent to each other along the X direction. In other words, the semiconductor chip **300** and the second passive element **202** may be positioned on adjacent (rather than opposite) sides of the first passive element **201**. In this case, the roughness of upper surfaces of a first sub-region R**2***a***-1** and a third sub-region R**2***c***-1** of a first passive element adjacent region R**2** may be less than the roughness of upper surfaces of a second sub-region R**2***b* and a fourth sub-region R**2***d* of the first passive element adjacent region R**2**. For example, the roughness of upper surfaces of the first sub-region R**2***a***-1** and the third sub-region R**2***c***-1** of the first passive element adjacent region R**2** may be less than the roughness of upper surfaces of the second sub-region R**2***b* and the fourth sub-region R**2***d* of the first passive element adjacent region R**2** by 80 nm or more.
(68) In some embodiments, the roughness of upper surfaces of a fifth sub-region Re**-1** to a seventh sub-region R**2***g***-1** of the first passive element adjacent region R**2** may be less than the roughness of the upper surface of an eighth sub-region R**2***h* of the first passive element adjacent region R**2**. For example, the roughness of upper surfaces of the fifth sub-region Re**-1** to the seventh sub-region R**2***g***-1** of the first passive element adjacent region R**2** may be less than the roughness of the upper surface of the eighth sub-region R**2***h* of the first passive element adjacent region R**2** by 80 nm or more.
(69) By forming the first sub-region R**2***a***-1**, the fifth sub-region R**2***e***-1**, and the sixth sub-region R**2***f***-1** of the first passive element adjacent region R**2** relatively flat, the flow of the first flux layer **271** to the first side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting, e.g., contacting, the second flux layer **272**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. **1**A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. **1**A) and the package substrate **100**.
(70) By forming the third sub-region R**2***c***-1**, the fifth sub-region R**2***e***-1**, and the seventh sub-region R**2***g***-1** of the first passive element adjacent region R**2** to be relatively flat, the first flux layer **271** may be prevented from contaminating the bump pad **124***a***-1** (refer to FIG. **1**A) under the semiconductor chip **300** positioned on the third side of the first passive element **201**.
(71) The first passive element **201** may be located on a second side of the second passive element **202**. In this case, the roughness of the upper surface of the second sub-region R**4***b***-1** of the second passive element adjacent region R**4** may be less than the roughness of upper surfaces of the first sub-region R**4***a*, the third sub-region R**4***c*, and the fourth sub-region R**4***d*. For example, the roughness of the upper surface of the second sub-region R**4***b***-1** of the second passive element adjacent region R**4** may be less than the roughness of upper surfaces of the first sub-region R**4***a*,

the third sub-region R**4***c*, and the fourth sub-region R**4***d* by about 80 nm or more.
(72) In some embodiments, the roughness of upper surfaces of a seventh sub-region R**4***g*-**1** and an eighth sub-region R**4***h*-**1** of the second passive element adjacent region R**4** may be less than the roughness of upper surfaces of a fifth sub-region R**4***e* and a sixth sub-region R**4***f* of the second passive element adjacent region R**4**. For example, the roughness of upper surfaces of the seventh sub-region R**4***g*-**1** and the eighth sub-region R**4***h*-**1** of the second passive element adjacent region R**4** may be less than the roughness of upper surfaces of the fifth sub-region R**4***e* and the sixth sub-region R**4***f* of the second passive element adjacent region R**4** by about 80 nm or more.
(73) By forming the second sub-region R**4***b*-**1**, the seventh sub-region R**4***g*-**1**, and the eighth sub-region R**4***h*-**1** of the second passive element adjacent region R**4** to be relatively flat, the flow of the second flux layer **272** to the second side of the second passive element **202** may be reduced. Accordingly, the second flux layer **272** may be prevented from meeting, e.g., contacting, the first flux layer **271**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. **1**A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. **1**A) and the package substrate **100**.
(74) FIG. **6** is a plan view showing a semiconductor package **1000***e* according to an embodiment. Hereinafter, differences between the semiconductor package **1000** shown in FIGS. **1**A to **1**C and the semiconductor package **1000***e* shown in FIG. **6** are described.
(75) Referring to FIG. **6**, the semiconductor package **1000***e* may include a first semiconductor chip **301** and a second semiconductor chip **302** on the remaining region R**3** of the package substrate **100** (instead of the single semiconductor chip **300** illustrated in FIGS. **1**A to **1**C). The first semiconductor chip **301** may be located on a first side of a passive element **200**, and the second semiconductor chip **302** may be located on a third side of the passive element **200**.
(76) In this case, the roughness of upper surfaces of a first sub-region R**2***a*-**1** and a third sub-region R**2***c*-**1** of the first passive element adjacent region R**2** may be less than the roughness of upper surfaces of a second sub-region R**2***b* and a fourth sub-region R**2***d* of the first passive element adjacent region R**2**. For example, the roughness of upper surfaces of the first sub-region R**2***a*-**1** and the third sub-region R**2***c*-**1** of the first passive element adjacent region R**2** may be less than the roughness of upper surfaces of the second sub-region R**2***b* and the fourth sub-region R**2***d* of the first passive element adjacent region R**2** by 80 nm or more.
(77) In some embodiments, the roughness of upper surfaces of a fifth sub-region R**2***e*-**1** to a seventh sub-region R**2***g*-**1** of the first passive element adjacent region R**2** may be less than the roughness of the upper surface of an eighth sub-region R**2***h* of the first passive element adjacent region R**2**. For example, the roughness of the upper surfaces of the fifth sub-region R**2***e*-**1** to the seventh sub-region R**2***g*-**1** of the first passive element adjacent region R**2** may be less than the roughness of the upper surface of the eighth sub-region R**2***h* of the first passive element adjacent region R**2** by 80 nm or more.
(78) By forming the first sub-region R**2***a*-**1**, the fifth sub-region R**2***e*-**1**, and the sixth sub-region R**2***f*-**1** of the first passive element adjacent region R**2** relatively flat, the flow of the first flux layer **271** to a first side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from contaminating the bump pad **124***a*-**1** (refer to FIG. **1**A) under the first semiconductor chip **301** positioned on the first side of the first passive element **201**.
(79) By forming the third sub-region R**2***c*-**1**, the fifth sub-region R**2***e*-**1**, and the seventh sub-region R**2***g*-**1** of the first passive element adjacent region R**2** to be relatively flat, the first flux layer **271** may be prevented from contaminating the bump pad **124***a*-**1** (refer to FIG. **1**A) under the second semiconductor chip **302** positioned on a third side of the first passive element **201**.
(80) FIG. **7** is a plan view showing a semiconductor package **1000***f* according to an embodiment. Hereinafter, differences between the semiconductor package **1000***d* shown in FIG. **5** and the semiconductor package **1000***f* shown in FIG. **7** are described.
(81) Referring to FIG. **7**, the semiconductor package **1000***f* may further include a third passive

element **203**, a third passive element connecting member **253** connecting the third passive element **203** to the package substrate **100**, and a third flux layer **273** adjacent to the third passive element **203** on the package substrate **100**.
(82) The second passive element **202** may be located on the first side of the first passive element **201**, and the third passive element **203** may be located on the third side of the first passive element **201**. In this case, the roughness of the upper surfaces of a first sub-region R**2***a*-**1** and a third sub-region R**2***c*-**1** of the first passive element adjacent region R**2** may be less than the roughness of upper surfaces of a second sub-region R**2***b* and a fourth sub-region R**2***d* of the first passive element adjacent region R**2**. For example, the roughness of upper surfaces of the first sub-region R**2***a*-**1** and the third sub-region R**2***c*-**1** of the first passive element adjacent region R**2** may be less than the roughness of upper surfaces of the second sub-region R**2***b* and the fourth sub-region R**2***d* of the first passive element adjacent region R**2** by 80 nm or more.
(83) In some embodiments, the roughness of upper surfaces of a fifth sub-region Re-**1** to a seventh sub-region R**2***g*-**1** of the first passive element adjacent region R**2** may be less than the roughness of the upper surface of an eighth sub-region R**2***h* of the first passive element adjacent region R**2**. For example, the roughness of upper surfaces of the fifth sub-region Re-**1** to the seventh sub-region R**2***g*-**1** of the first passive element adjacent region R**2** may be less than the roughness of the upper surface of the eighth sub-region R**2***h* of the first passive element adjacent region R**2** by 80 nm or more.
(84) By forming the first sub-region R**2***a*-**1**, the fifth sub-region R**2***e*-**1**, and the sixth sub-region R**2***f*-**1** of the first passive element adjacent region R**2** relatively flat, the flow of the first flux layer **271** to the first side of the first passive element **201**, e.g., along the X direction, may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting, e.g., contacting, the second flux layer **272**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. **1**A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. **1**A) and the package substrate **100**.
(85) By forming the third sub-region R**2***c*-**1**, the fifth sub-region R**2***e*-**1**, and the seventh sub-region R**2***g*-**1** of the first passive element adjacent region R**2** to be relatively flat, the flow of the first flux layer **271** to the third side of the first passive element **201**, e.g., along the Y direction, may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting, e.g., contacting, the third flux layer **273**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. **1**A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. **1**A) and the package substrate **100**.
(86) FIG. **8** is a plan view showing a semiconductor package **1000***g* according to an embodiment. Hereinafter, differences between the semiconductor package **1000***c* shown in FIG. **4** and the semiconductor package **1000***g* shown in FIG. **8** are described.
(87) Referring to FIG. **8**, the second passive element **202** may be positioned on the second side of the first passive element **201**, and the semiconductor chip **300** may be positioned on the first side of the first passive element **201**. In this case, the roughness of upper surfaces of a first sub-region R**2***a*-**1** and a second sub-region R**2***b*-**1** of a first passive element adjacent region R**2** may be less than the roughness of upper surfaces of a third sub-region R**2***c* and a fourth sub-region R**2***d* of the first passive element adjacent region R**2**. For example, the roughness of upper surfaces of the first sub-region R**2***a*-**1** and the second sub-region R**2***b*-**1** of the first passive element adjacent region R**2** may be less than the roughness of upper surfaces of the third sub-region R**2***c* and the fourth sub-region R**2***d* of the first passive element adjacent region R**2** by 80 nm or more.
(88) The roughness of upper surfaces of a fifth sub-region R**2***e*-**1** to an eighth sub-region R**2***h*-**1** of the first passive element adjacent region R**2** may be less than the roughness of upper surfaces of a third sub-region R**2***c* and a fourth sub-region R**2***d* of the first passive element adjacent region R**2**. For example, the roughness of upper surfaces of the fifth sub-region R**2***e*-**1** to the eighth sub-region R**2***h*-**1** of the first passive element adjacent region R**2** may be less than the roughness of upper

surfaces of the third sub-region R**2***c* and the fourth sub-region R**2***d* of the first passive element adjacent region R**2** by 80 nm or more.
(89) By forming the first sub-region R**2***a***-1**, the fifth sub-region R**2***e***-1**, and the sixth sub-region R**2***f***-1** of the first passive element adjacent region R**2** relatively flat, the flow of the first flux layer **271** to the first side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from contaminating the bump pad **124***a***-1** (refer to FIG. **1**A) under the semiconductor chip **300** positioned on the first side of the first passive element **201**.
(90) By forming the second sub-region R**2***b***-1**, the seventh sub-region R**2***g***-1**, and the eighth sub-region R**2***h***-1** of the first passive element adjacent region R**2** to be relatively flat, the flow of the first flux layer **271** to the second side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting, e.g., contacting, the second flux layer **272**. Therefore, an increase in the concentration of the flux mixed with the sealing portion may be prevented (see **400**, FIG. **1**A). Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. **1**A) may be prevented.
(91) FIG. **9** is a plan view showing a semiconductor package **1000***h* according to an embodiment. Hereinafter, differences between the semiconductor package **1000***g* shown in FIG. **8** and the semiconductor package **1000***h* shown in FIG. **9** are described.
(92) Referring to FIG. **9**, the semiconductor package **1000***h* may include the passive element **200**, the first semiconductor chip **301**, and the second semiconductor chip **302**, instead of the first passive element **201**, the second passive element **202**, and the semiconductor chip **300**. The first semiconductor chip **301** may be located on a second side of the passive element **200**, and the second semiconductor chip **302** may be located on a first side of the passive element **200**. In other words, the passive element **200** may be between the first and second semiconductor chips **301** and **302** along the X direction.
(93) By forming a first sub-region R**2***a***-1**, a fifth sub-region R**2***e***-1**, and a sixth sub-region R**2***f***-1** of a passive element adjacent region R**2** relatively flat, the flow of the flux layer **270** to the first side of the passive element **200** may be reduced. Accordingly, the flux layer **270** may be prevented from contaminating the bump pad **124***a***-1** (refer to FIG. **1**A) under the second semiconductor chip **302** positioned on the first side of the passive element **200**.
(94) By forming a second sub-region R**2***b***-1**, a seventh sub-region R**2***g***-1**, and an eighth sub-region R**2***h***-1** of the passive element adjacent region R**2** to be relatively flat, the flow of the flux layer **270** to the second side of the passive element **200** may be reduced. Accordingly, the flux layer **270** may be prevented from contaminating the bump pad **124***a***-1** (refer to FIG. **1**A) under the first semiconductor chip **301** positioned on the second side of the passive element **200**.
(95) FIG. **10** is a plan view showing a semiconductor package **1000***i* according to an embodiment. Hereinafter, differences between the semiconductor package **1000***g* shown in FIG. **8** and the semiconductor package **1000***i* shown in FIG. **10** are described.
(96) Referring to FIG. **10**, the semiconductor package **1000***i* may further include the third passive element **203** on the package substrate **100**, the third passive element connecting member **253** connecting the third passive element **203** to the package substrate **100**, and the third flux layer **273** adjacent to the third passive element **203** on the package substrate **100**. The second passive element **202** may be positioned on the second side of the first passive element **201**, and the third passive element **203** may be positioned on the first side of the first passive element **201**.
(97) By forming a first sub-region R**2***a***-1**, the fifth sub-region R**2***e***-1**, and a sixth sub-region R**2***f***-1** of the first passive element adjacent region R**2** relatively flat, the flow of the first flux layer **271** to the first side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting, e.g., contacting, the third flux layer **273**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. **1**A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. **1**A) and the package substrate **100**.

(98) By forming a second sub-region R**2***b*-**1**, a seventh sub-region R**2***g*-**1**, and an eighth sub-region R**2***h*-**1** of the first passive element adjacent region R**2** to be relatively flat, the flow of the first flux layer **271** to the second side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting, e.g., contacting, the second flux layer **272**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. **1**A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. **1**A) and the package substrate **100**.
(99) FIG. **11** is a plan view showing a semiconductor package **1000***j* according to an embodiment. Hereinafter, differences between the semiconductor package **1000** shown in FIGS. **1**A to **1**C and the semiconductor package **1000***j* shown in FIG. **11** are described.
(100) Referring to FIG. **11**, the semiconductor package **1000***j* may include the first semiconductor chip **301**, the second semiconductor chip **302**, and a third semiconductor chip **303** (instead of the single semiconductor chip **300** in FIGS. **1**A-**1**C). The first semiconductor chip **301** may be located on the second side of a passive element **200**, the second semiconductor chip **302** may be located on the first side of the passive element **200**, and the third semiconductor chip **303** may be located on the third side of the passive element **200**.
(101) In this case, the roughness of top surfaces of a first sub-region R**2***a*-**1** to a third sub-region R**2***c*-**1** of the passive element adjacent region R**2** may be less than the roughness of the upper surface of a fourth sub-region R**2***d* of the passive element adjacent region R**2**. For example, the roughness of upper surfaces of the first sub-region R**2***a*-**1** to the third sub-region R**2***c*-**1** of the passive element adjacent region R**2** may be less than the roughness of the upper surface of the fourth sub-region R**2***d* of the passive element adjacent region R**2** by about 80 nm or more.
(102) The roughness of the upper surfaces of a fifth sub-region R**2***e*-**1** to an eighth sub-region R**2***h*-**1** of the passive element adjacent region R**2** may be less than the roughness of the upper surface of a fourth sub-region R**2***d* of the passive element adjacent region R**2**. For example, the roughness of the upper surfaces of the fifth sub-region R**2***e*-**1** to the eighth sub-region R**2***h*-**1** of the passive element adjacent region R**2** may be less than the roughness of the upper surface of the fourth sub-region R**2***d* of the passive element adjacent region R**2** by about 80 nm or more.
(103) By forming the first sub-region R**2***a*-**1**, the fifth sub-region R**2***e*-**1**, and the sixth sub-region R**2***f*-**1** of the passive element adjacent region R**2** relatively flat, the flow of the flux layer **270** to the first side of the passive element **200** may be reduced. Therefore, the flux layer **270** may be prevented from contaminating the bump pad **124***a*-**1** (refer to FIG. **1**A) under the second semiconductor chip **302** positioned on the first side of the passive element **200**.
(104) By forming the second sub-region R**2***b*-**1**, the seventh sub-region R**2***g*-**1**, and the eighth sub-region R**2***h*-**1** of the passive element adjacent region R**2** to be relatively flat, the flow of the flux layer **270** to the second side of the passive element **200** may be reduced. Accordingly, the flux layer **270** may be prevented from contaminating the bump pad **124***a*-**1** (refer to FIG. **1**A) under the first semiconductor chip **301** positioned on the second side of the passive element **200**.
(105) By forming the third sub-region R**2***c*-**1**, the fifth sub-region R**2***e*-**1**, and the seventh sub-region R**2***g*-**1** of the passive element adjacent region R**2** to be relatively flat, the flow of the flux layer **270** to the third side of the passive element **200** may be reduced. Accordingly, the flux layer **270** may be prevented from contaminating the bump pad **124***a*-**1** (refer to FIG. **1**A) under the third semiconductor chip **303** positioned on the third side of the passive element **200**.
(106) FIG. **12** is a plan view showing a semiconductor package **1000***k* according to an embodiment. Hereinafter, differences between the semiconductor package **1000***j* shown in FIG. **11** and the semiconductor package **1000***k* shown in FIG. **12** are described.
(107) Referring to FIG. **12**, the semiconductor package **1000***k* may include the first passive element **201** and the second passive element **202** on the package substrate **100** (instead of the single passive element **200** and the third semiconductor chip **303** in FIG. **11**). The semiconductor package **1000***k* may further include the second passive element connecting member **252** that connects the second

passive element **202** to the package substrate **100** and the second flux layer **272** adjacent the second passive element **202** on the package substrate **100**.

(108) By forming a first sub-region R**2***a*-**1**, a fifth sub-region R**2***e*-**1**, and a sixth sub-region R**2***f*-**1** of a first passive element adjacent region R**2** relatively flat, the flow of a first flux layer **271** to a first side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from contaminating the bump pads **124***a*-**1** (refer to FIG. **1**A) under the second semiconductor chip **302** positioned on the first side of the first passive element **201**.

(109) By forming a second sub-region R**2***b*-**1**, a seventh sub-region R**2***g*-**1**, and an eighth sub-region R**2***h*-**1** of the first passive element adjacent region R**2** to be relatively flat, the flow of the first flux layer **271** to a second side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from contaminating the bump pad **124***a*-**1** (refer to FIG. **1**A) under the first semiconductor chip **301** positioned on the second side of the first passive element **201**.

(110) By forming a third sub-region R**2***c*-**1**, the fifth sub-region R**2***e*-**1**, and a seventh sub-region R**2***g*-**1** of the first passive element adjacent region R**2** to be relatively flat, the flow of the first flux layer **271** to the third side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting the second flux layer **272**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. **1**A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. **1**A) and the package substrate **100**.

(111) FIG. **13** is a plan view showing a semiconductor package **10001** according to an embodiment. Hereinafter, differences between the semiconductor package **1000***k* shown in FIG. **12** and the semiconductor package **10001** shown in FIG. **13** are described.

(112) Referring to FIG. **13**, the semiconductor package **10001** may include the third passive element **203** and a single semiconductor chip **300** (instead of the first semiconductor chip **301** and the second semiconductor chip **302** in FIG. **12**). The semiconductor package **10001** may further include the third passive element connecting member **253** connecting the third passive element **203** to the package substrate **100** and the third flux layer **273** adjacent to the third passive element **203** on the package substrate **100**. The second passive element **202** may be located on the second side of the first passive element **201**, the third passive element **203** may be located on the first side of the first passive element **201**, and the semiconductor chip **300** may be located on a third side of the first passive element **201**.

(113) By forming a first sub-region R**2***a*-**1**, a fifth sub-region R**2***e*-**1**, and a sixth sub-region R**2***f*-**1** of the first passive element adjacent region R**2** relatively flat, the flow of the first flux layer **271** to the first side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting the third flux layer **273**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. **1**A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. **1**A) and the package substrate **100**.

(114) By forming a second sub-region R**2***b*-**1**, a seventh sub-region R**2***g*-**1**, and an eighth sub-region R**2***h*-**1** of the first passive element adjacent region R**2** to be relatively flat, the flow of the first flux layer **271** to the second side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting the second flux layer **272**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. **1**A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. **1**A) and the package substrate **100**.

(115) By forming a third sub-region R**2***c*-**1**, a fifth sub-region R**2***e*-**1**, and a seventh sub-region R**2***g*-**1** of the first passive element adjacent region R**2** to be relatively flat, the flow of the first flux layer **271** to the third side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from contaminating the bump pad **124***a*-**1** (refer to FIG. **1**A) under the

semiconductor chip **300** positioned on the third side of the first passive element **201**.
(116) FIG. **14** is a plan view showing a semiconductor package **1000***m* according to an embodiment. Hereinafter, differences between the semiconductor package **10001** shown in FIG. **13** and the semiconductor package **1000***m* shown in FIG. **14** are described.
(117) Referring to FIG. **14**, the semiconductor package **1000***m* may further include a fourth passive element **204** on the package substrate **100**, a fourth passive element connecting member **254** connecting the fourth passive element **204** to the package substrate **100**, and a fourth flux layer **274** adjacent to the fourth passive element **204** on the package substrate **100**. The second passive element **202** may be located on the second side of the first passive element **201**, the third passive element **203** may be located on the first side of the first passive element **201**, and the fourth passive element **204** may be located on the third side of the second passive element **202**.
(118) By forming a first sub-region R**2***a*-**1**, a fifth sub-region R**2***e*-**1**, and a sixth sub-region R**2***f*-**1** of the first passive element adjacent region R**2** relatively flat, the flow of the first flux layer **271** to the first side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting the third flux layer **273**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. **1**A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. **1**A) and the package substrate **100**.
(119) By forming a second sub-region R**2***b*-**1**, a seventh sub-region R**2***g*-**1**, and an eighth sub-region R**2***h*-**1** of the first passive element adjacent region R**2** to be relatively flat, the flow of the first flux layer **271** to the second side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting the second flux layer **272**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. **1**A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. **1**A) and the package substrate **100**.
(120) By forming a third sub-region R**2***c*-**1**, a fifth sub-region R**2***e*-**1**, and a seventh sub-region R**2***g*-**1** of the first passive element adjacent region R**2** to be relatively flat, the flow of the first flux layer **271** to a third side of the first passive element **201** may be reduced. Accordingly, the first flux layer **271** may be prevented from meeting the fourth flux layer **274**. Therefore, an increase in the concentration of the flux mixed with the sealing portion **400** (refer to FIG. **1**A) may be prevented. Accordingly, it is possible to prevent delamination between the sealing portion **400** (refer to FIG. **1**A) and the package substrate **100**.
(121) FIG. **15** is a cross-sectional view illustrating a semiconductor package **1000***n* according to an embodiment. Hereinafter, differences between the semiconductor packages **1000** shown in FIGS. **1**A to **1**C and the semiconductor package **1000***n* shown in FIG. **15** are described.
(122) Referring to FIG. **15**, the semiconductor package **1000***n* may include a wire **370** instead of the chip bump **350**. Also, the package substrate **100** may include a wire pad **124***a*-**3** instead of the bump pad **124***a*-**1**. The wire **370** may connect the semiconductor chip **300** to the wire pad **124***a*-**3** of the package substrate **100**.
(123) By forming the remaining region R**3** to be relatively flat, the flow of a flux layer **270** onto the remaining region R**3** may be reduced. Accordingly, the flux layer **270** may be prevented from contaminating the wire pads **124***a*-**3**.
(124) FIG. **16** is a cross-sectional view illustrating a semiconductor package **10000** according to an embodiment; Hereinafter, differences between the semiconductor package **1000***k* shown in FIGS. **1**A to **12** and the semiconductor package **10000** shown in FIG. **16** are described.
(125) Referring to FIG. **16**, the flux layer **270** may not be present on the passive element region R**1** and the passive element adjacent region R**2**. For example, after the passive element **200** is mounted on the package substrate **100**, the flux layer **270** may be removed. For example, when the flux layer **270** includes an aqueous flux, the flux layer **270** may be removed using water.
(126) FIG. **17** is a cross-sectional view illustrating a semiconductor package **1000***p* according to an

embodiment. Hereinafter, differences between the semiconductor package **1000** shown in FIGS. **1**A to **1**C and the semiconductor package **1000***p* shown in FIG. **17** are described.
(127) Referring to FIG. **17**, the semiconductor package **1000***p* may further include a chip flux layer **370-1**. When the semiconductor chip **300** is mounted on the package substrate **100**, the chip flux layer **370-1** around the chip bump **350** is generally used. After the semiconductor chip **300** is mounted on the package substrate **100**, the chip flux layer **370-1** may be removed. For example, when the chip flux layer **370-1** includes an aqueous flux, the chip flux layer **370-1** may be removed using water. However, in another embodiment, as shown in FIG. **17**, the chip plus layer **370-1** may remain without being removed.
(128) FIG. **18** is a cross-sectional view illustrating a semiconductor package **1000***q* according to an embodiment. Hereinafter, differences between the semiconductor package **1000** shown in FIGS. **1**A to **1**C and the semiconductor package **1000***q* shown in FIG. **18** are described.
(129) Referring to FIG. **18**, the semiconductor package **1000***q* may include the first semiconductor chip **301** and the second semiconductor chip **302** on the package substrate **100** instead of the semiconductor chip **300**. In some embodiments, the first semiconductor chip **301** may be a memory controller chip, and the second semiconductor chip **302** may be a memory chip, e.g., a flash memory chip. The first semiconductor chip **301** may be connected to the bump pads **124***a***-1** of the package substrate **100** using the chip bumps **350**. The package substrate **100** may further include the wire pads **124***a***-3** on the fifth insulating layer **121***e*. The semiconductor package **1000***q* may further include the wire **370** connecting the second semiconductor chip **302** to the wire pad **124***a***-3** of the package substrate **100**.
(130) Although FIG. **18** illustrates that the semiconductor package **1000***q* includes two semiconductor chips, e.g., the first semiconductor chip **301** and the second semiconductor chip **302**, the number of semiconductor chips included in the semiconductor package **1000***q* is not limited to two. For example, the semiconductor package **1000***q* may include a plurality of second semiconductor chips **302**. The plurality of second semiconductor chips **302** may be a plurality of flash memory chips. The plurality of second semiconductor chips **302** may be stacked on the package substrate **100**. Additionally, or alternatively, the semiconductor package **1000***q* may further include a third semiconductor chip, e.g., a DRAM chip, located on, e.g., the package substrate **100**, the first semiconductor chip **301**, or the second semiconductor chip **302**.
(131) By forming the first semiconductor chip **301** and the second semiconductor chip **302** on the remaining region R**3** having a relatively flat upper surface, the flux layer **270** may be prevented from contaminating the bump pad **124***a***-1** and the wire pad **124***a***-3**. Further, a flux layer used to mount the first semiconductor chip **301** on the package substrate **100** may be prevented from contaminating the wire pads **124***a***-3**.
(132) FIG. **19** is a cross-sectional view illustrating a semiconductor package **1000***r* according to an embodiment. Hereinafter, differences between the semiconductor package **1000***q* shown in FIG. **18** and the semiconductor package **1000***r* shown in FIG. **19** are described.
(133) Referring to FIG. **19**, the second semiconductor chip **302** may be stacked on the first semiconductor chip **301**. Although FIG. **19** illustrates that the semiconductor package **1000***r* includes two semiconductor chips, e.g., the first semiconductor chip **301** and the second semiconductor chip **302**, the number of semiconductor chips included in the semiconductor package **1000***r* is not limited to two. For example, the semiconductor package **1000***r* may include a plurality of second semiconductor chips **302**. The plurality of second semiconductor chips **302** may be a plurality of flash memory chips. The plurality of second semiconductor chips **302** may be stacked on the first semiconductor chip **301**. Additionally, or alternatively, the semiconductor package **1000***q* may further include a third semiconductor chip, e.g., a DRAM chip, positioned between the first semiconductor chip **301** and the second semiconductor chip **302** or on the second semiconductor chip **302**.
(134) FIG. **20**A is a cross-sectional view illustrating a semiconductor package **1000***s* according to

an embodiment. FIG. **20**B is an enlarged view of region B**2** of FIG. **20**A. Hereinafter, differences between the semiconductor package **1000***n* shown in FIG. **15** and the semiconductor package **1000***s* shown in FIGS. **20**A and **20**B are described.

(135) Referring to FIGS. **20**A and **20**B, the semiconductor package **1000***s* may include a wire pad adjacent region R-**1** adjacent to the wire pad **124***a*-**3** and a remaining region R-**2**. The semiconductor chip **300** may be located on the remaining region R-**2**. The roughness of the upper surface of the wire pad adjacent region R-**1** may be less than the roughness of the upper surface of the remaining region R-**2**. Accordingly, the flux layer **270** may be prevented from flowing into the wire pad adjacent region R-**1** and contaminating the wire pad **124***a*-**3**.

(136) FIG. **21** is a flowchart illustrating a method **2000** of manufacturing a semiconductor package according to an embodiment.

(137) Referring to FIGS. **21** and **1**A to **1**C, the package substrate **100** may be roughened in operation S**2100**. Through the operation S**2100**, the roughness of the upper surface of at least one of the first sub-regions R**2***a* to R**2***d* may be greater than the roughness of the upper surface of a remaining region R**3**.

(138) For example, a mask may be arranged on a region of the package substrate **100** that should be relatively flat, and the region of the package substrate **100** that should be relatively rough may be etched. For etching, dry etching, wet etching, or a combination thereof may be used. For example, plasma etching may be used. For example, a mask may be arranged on the remaining region R**3**, and the passive element region R**1** and the passive element adjacent region R**2** may be etched. The etched passive element region R**1** and passive element adjacent region R**2** may be relatively rough, and a non-etched remaining region R**3** may be relatively flat. Alternatively, a mask may be arranged on the remaining region R**3** and the passive element region R**1**, and the passive element adjacent region R**2** may be etched. As shown in FIG. **2**, the etched passive adjacent region R**2** may be relatively rough, and the non-etched remaining region R**3** and passive element region R**1** may be relatively flat. Alternatively, referring to FIGS. **20**A and **20**B, a mask may be arranged on the wire pad adjacent region R-**1**, and the remaining region R-**2** may be etched. The etched remaining region R-**2** may be relatively rough, and a non-etched wire pad adjacent region R-**1** may be relatively flat.

(139) Additionally, a mask may be arranged on a region of the package substrate **100** that should be relatively rough, and the region of the package substrate **100** that should be relatively flat may be further etched. However, by adjusting the etching conditions, e.g., time, concentration, power, flow rate, pressure, temperature, etc., the etching may be performed weaker than the first etching. For example, a mask may be arranged on the passive element region R**1** and the passive element region R**2**, and the remaining region R**3** may be lightly etched. Alternatively, a mask may be arranged on the passive element region R**2**, and the passive element region R**1** and the remaining region R**3** may be lightly etched. Alternatively, a mask may be arranged on the remaining region R-**2**, and the wire pad adjacent region R-**1** may be lightly etched.

(140) Next, the passive element **200** may be mounted on the passive element region R**1** of the package substrate **100** in operation S**2200**. A solder paste may be used to mount the passive element **200** on the passive element region R**1** of the package substrate **100**. The passive element connecting member **250** and the flux layer **270** may be formed of the solder paste. The flux layer **270** may not be removed as shown in FIG. **1**A or may be removed as shown in FIG. **16**.

(141) Next, the semiconductor chip **300** is mounted on the remaining region R**3** of the package substrate **100** in operation S**2300**. Solder balls and flux may be used to mount the semiconductor chip **300** on the remaining region R**3** of the package substrate **100**. The chip bump **350** may be formed of the solder ball. The flux may be removed as shown in FIG. **1**A, or may not be removed as shown in FIG. **17**. As shown in FIG. **15**, alternatively, the semiconductor chip **300** may be connected to the remaining region R**3** of the package substrate **100** using the wire **370**.

(142) Next, the sealing portion **400** covering the package substrate **100**, the passive element **200**, and the semiconductor chip **300** may be formed in operation S**2400**. Finally, the external

connection terminal **500** may be attached by attaching a solder ball to the lower surface of the package substrate **100** in operation S**2500**.

(143) By way of summation and review, when a flux layer on a package substrate is mixed with a sealing portion, delamination may occur between the package substrate and the sealing portion. In addition, the flux layer may contaminate a bump pad of the package substrate and a chip bump or a wire pad of the package substrate.

(144) In contrast, embodiments provide a semiconductor package for preventing delamination between a package substrate and a sealing portion and preventing contamination of a bump pad or a wire pad by a flux layer. That is, the roughness of the upper surface of the region adjacent to the passive element may be greater than the roughness of the upper surface of the remaining regions, thereby allowing the flux layer to spread more widely over the relatively rough surface. Accordingly, the concentration of the flux layer mixed in the sealing portion can be reduced and delamination between the package substrate and the sealing portion may be minimized. Also, because the upper surface of the remaining region is relatively flat, it may be difficult for the flux layer to flow into the remaining region, thereby preventing the flux layer from contaminating the bump pad or the wire pad.

(145) Example embodiments have been disclosed herein, and although specific terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent to one of ordinary skill in the art as of the filing of the present application, features, characteristics, and/or elements described in connection with a particular embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present invention as set forth in the following claims.

Claims

1. A semiconductor package, comprising: a package substrate including a passive element region, a peripheral region adjacent to the passive element region, and a remaining region; a first passive element on a non-conductive upper surface of the package substrate in the passive element region; a passive element connecting member connecting the first passive element to the package substrate; a first semiconductor chip on a non-conductive upper surface of the package substrate in the remaining region; and a sealing portion covering the package substrate, the first passive element, and the first semiconductor chip, wherein the peripheral region includes a first sub-region on a first side of the first passive element, a second sub-region on a second side opposite the first side of the first passive element, a third sub-region on a third side of the first passive element, and a fourth sub-region on a fourth side opposite the third side of the first passive element, wherein a roughness of a non-conductive upper surface of the package substrate in at least one of the first sub-region to the fourth sub-region is greater than a roughness of the non-conductive upper surface of the package substrate in the remaining region, and wherein the first semiconductor chip overlaps the remaining region in a plan view.
2. The semiconductor package as claimed in claim 1, wherein the roughness of the non-conductive upper surface of the package substrate in the passive element region is greater than the roughness of the non-conductive upper surface of the package substrate in the remaining region.
3. The semiconductor package as claimed in claim 1, wherein the roughness of the non-conductive upper surface of the package substrate in the passive element region is less than the roughness of the non conductive upper surface of the package substrate in the at least one of the first sub-region to the fourth sub-region.
4. The semiconductor package as claimed in claim 1, wherein the roughness of the non-conductive

upper surface of the package substrate in the first sub-region is less than the roughness of the non-conductive upper surface of the package substrate in each of the second sub-region to the fourth sub-region.

5. The semiconductor package as claimed in claim 4, wherein the first semiconductor chip is positioned on the first side of the first passive element.

6. The semiconductor package as claimed in claim 4, further comprising a second passive element positioned on the first side of the first passive element.

7. The semiconductor package as claimed in claim 1, wherein the roughness of the non-conductive upper surface of the package substrate in each of the first sub-region and the third sub-region is less than the roughness of the non-conductive upper surface of the package substrate in each of the second sub-region and the fourth sub-region.

8. The semiconductor package as claimed in claim 7, wherein: the first semiconductor chip is positioned on one of the first side and the third side of the first passive element, and a second passive element or a second semiconductor chip is positioned on the other of the first side and the third side of the first passive element.

9. The semiconductor package as claimed in claim 7, further comprising a second passive element and a third passive element respectively positioned on the first side and the third side of the first passive element.

10. The semiconductor package as claimed in claim 1, wherein the roughness of the non-conductive upper surface of the package substrate in each of the first sub-region and the second sub-region is less than the roughness of the non-conductive upper surface of the package substrate in each of the third sub-region and the fourth sub-region.

11. The semiconductor package as claimed in claim 10, wherein: the first semiconductor chip is positioned on one of the first side and the second side of the first passive element, and a second passive element or a second semiconductor chip is positioned on the other one of the first side and the second side of the first passive element.

12. The semiconductor package as claimed in claim 1, wherein the roughness of the non-conductive upper surface of the package substrate in each of the first sub-region, the second sub-region, and the third sub-region is less than the roughness of the non-conductive upper surface of the package substrate in the fourth sub-region.

13. The semiconductor package as claimed in claim 12, wherein the first semiconductor chip is on one of the first side, the second side, and the third side of the first passive element.

14. The semiconductor package as claimed in claim 13, further comprising: a second semiconductor chip and a third semiconductor chip respectively positioned on the other two sides of the first side, the second side, and the third side of the first passive element, a second semiconductor chip and a second passive element respectively positioned on the other two sides of the first side, the second side, and the third side of the first passive element, or a second passive element and a third passive element respectively positioned on the other two sides of the first side, the second side, and the third side of the first passive element.

15. The semiconductor package as claimed in claim 1, wherein the passive element connecting member is disposed in a recess in the package substrate and contacts sidewalls of the recess.

16. The semiconductor package as claimed in claim 1, further comprising a flux layer disposed on the non-conductive upper surface of the package substrate in at least the second sub-region of the peripheral region and contacting a side surface of the passive element connecting member.

17. A semiconductor package, comprising: a package substrate including a wire pad, a peripheral region adjacent to the wire pad, and a remaining region; a passive element on a non-conductive upper surface of the package substrate in the remaining region; a passive element connecting member connecting the passive element to the package substrate; a semiconductor chip on the non-conductive upper surface of the package substrate in the remaining region; a wire electrically connecting the semiconductor chip to the wire pad; and a sealing portion covering the package

substrate, the passive element, the semiconductor chip, and the wire, and contacting a top horizontal surface of the passive element connecting member, wherein a roughness of a non-conductive upper surface of the package substrate in the peripheral region is less than a roughness of the non-conductive upper surface of the substrate in the remaining region, and wherein the semiconductor chip overlaps the remaining region in a plan view.

18. A semiconductor package, comprising: a package substrate including a passive element region, a peripheral region adjacent to the passive element region, and a remaining region; a plurality of external connection terminals on a lower surface of the package substrate; a passive element on a non-conductive upper surface of the package substrate in the passive element region; a passive element connecting member connecting the passive element to the package substrate, a memory controller chip on a non-conductive upper surface of the package substrate in the remaining region; a chip bump connecting the memory controller chip to the package substrate; a memory chip on the memory controller chip or the remaining region; a wire connecting the memory chip to the package substrate; a sealing portion covering the package substrate, the passive element, the memory controller chip, the chip bump, the memory chip, and the wire; and a flux layer disposed between the package substrate a ad the passive element, the flux layer contacting a side surface of the passive element connecting member, wherein the peripheral region includes a first sub-region on a first side of the passive element, a second sub-region on a second side opposite the first side of the passive element, a third sub-region on a third side of the passive element, and a fourth sub-region on a fourth side opposite the third side of the passive element, wherein a roughness of a non-conductive upper surface of the package substrate in at least one of the first sub-region to the fourth sub-region is greater than a roughness of the non-conductive upper surface of the package substrate in the remaining region, wherein the memory controller chip overlaps the non-conductive upper surface of the package substrate in the remaining region in plan view, and wherein the flux layer is disposed on the non-conductive upper surface of the package substrate in the second sub-region of the peripheral region.

19. The semiconductor package as claimed in claim 18, wherein a width of the at least one of the first sub-region to the fourth sub-region along a direction perpendicular to a thickness direction of the package substrate is about 10 μm to about 500 μm.

20. The semiconductor package as claimed in claim 18, wherein a difference between the roughness of the non-conductive upper surface of the package substrate in the at least one of the first sub-region to the fourth sub-region and the roughness of the non-conductive upper surface of the package substrate in the remaining regions is 80 nm or more, and wherein the sealing portion contacts a top horizontal surface of the passive element connecting member.